Finding and Verifying Inverse Functions

Algebra 2 Composition, Inverses and Transformations

Finding an inverse by solving for the variable, verifying a pair by composition, and using inverse values and domains

Directions. Show the work, not just the result.

- **To find an inverse:** write $y = f(x)$, swap x and y, solve the new equation for y, and rename the result $f^{-1}(x)$.
- **To verify a pair:** show *both* $f(g(x)) = x$ and $g(f(x)) = x$. One direction alone is not enough.
- f^{-1} means the inverse function, never $\dfrac{1}{f}$.

1. Find the inverse of $f(x) = x + 7$, then check your answer by substituting it into f.

Answer: ________________

2. Find the inverse of $f(x) = 5x$, then check it by composing f with the function you found.

Answer: ________________

3. Find $f^{-1}(x)$ for $f(x) = 3x - 8$. Show the swap-and-solve steps, and verify that $f(f^{-1}(x)) = x$.

Answer: ________________

4. Let $f(x) = 2x + 9$ and $g(x) = \dfrac{x-9}{2}$. Verify that these two are inverses by computing *both* $f(g(x))$ and $g(f(x))$ and simplifying each one.

Answer: ________________

5. For $f(x) = 4x - 1$, find $f^{-1}(11)$. Do it by first writing a formula for $f^{-1}(x)$, then evaluating that formula at 11.

Answer: ________________

6. Find the inverse of $f(x) = \dfrac{x-4}{5}$. Check your answer by simplifying $f^{-1}(f(x))$.

Answer: ________________

7. Find the inverse of $f(x) = \sqrt{x-2}$, whose domain is $x \geq 2$. What is the domain of the inverse, and why is it not all real numbers?

Answer: ________________

8. Sofia claims $f(x) = \dfrac{x+6}{2}$ and $g(x) = 2x - 6$ are inverses. Test the claim by simplifying both $f(g(x))$ and $g(f(x))$, and say whether she is right.

Answer: ________________

9. Let $f(x) = x^2$ with the domain restricted to $x \geq 0$. Write $f^{-1}(x)$ and use it to find $f^{-1}(49)$. Why does the restriction matter here?

Answer: ________________

10. Find the inverse of the rational function $f(x) = \dfrac{2x+1}{x-3}$. Swap the variables, then collect every y term on one side and factor before dividing.

Answer: ________________

11. Let $f(x) = 3x + 5$. Find $f^{-1}(-4)$ two different ways: once from a formula for $f^{-1}(x)$, and once by solving $f(x) = -4$ directly. The two answers must agree.

Answer: ________________

12. Let $f(x) = x^2 - 4$.
(a) Explain in a sentence why f has no inverse function on all of the real numbers. **(b)** Restrict the domain to $x \geq 0$, write $f^{-1}(x)$ for that restricted function, and use it to find $f^{-1}(5)$.

Answer: ____________________